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Semiconductor device including stack structure and trenches

Abstract

A semiconductor device includes a plurality of blocks on a substrate. Trenches are disposed between the plurality of blocks. Conductive patterns are formed inside the trenches. A lower end of an outermost trench among the trenches is formed at a level higher than a level of a lower end of the trench adjacent to the outermost trench. Each of the blocks includes insulating layers and gate electrodes, which are alternately and repeatedly stacked. Pillars pass through the insulating layers and the gate electrodes along a direction orthogonal to an upper surface of the substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This U.S. non-provisional patent application is a continuation of U.S. patent application Ser. No. 16/890,400 filed on Jun. 2, 2020, which is a continuation of U.S. patent application Ser. No. 15/941,800 filed on Mar. 30, 2018, now U.S. Pat. No. 10,707,229 issued on Jul. 7, 2020, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2017-0106033, filed on Aug. 22, 2017, in the Korean Intellectual Property Office (KIPO), the disclosures of which are incorporated by reference herein in their entireties.

1. TECHNICAL FIELD
(1) Exemplary embodiments of the present inventive concept relate to a semiconductor device, and more particularly to a semiconductor device including a stack structure and trenches.
2. DISCUSSION OF RELATED ART
(2) A degree of integration of semiconductor devices may be increased by the lighting, thinning, and shortening of electronic devices. As an example, a technique using a stack structure in which insulating layers and electrode layers are alternately and repeatedly stacked on a substrate has been attempted. A plurality of cell blocks may be defined by a plurality of trenches which vertically pass through the stack structure and are in parallel. Each of the trenches may have a relatively high aspect ratio. In a semiconductor device including trenches, an outermost trench among the trenches may be vulnerable to various defects caused by mold leaning.
SUMMARY
(3) An exemplary embodiment of the present inventive concept provides a semiconductor device having a reduced number of defects occurring at edges of a cell array region and a relatively high degree of integration.
(4) An exemplary embodiment of the present inventive concept provides a method of forming a semiconductor device with a reduced number of defects occurring at edges of a cell array region and a relatively high degree of integration.
(5) A semiconductor device according to an exemplary embodiment of the present inventive concept includes a plurality of blocks on a substrate. Trenches are disposed between the plurality of blocks. Conductive patterns are formed inside the trenches. A lower end of an outermost trench among the trenches is formed at a level higher than a level of a lower end of the trench adjacent to the outermost trench. Each of the blocks includes insulating layers and gate electrodes, which are alternately and repeatedly stacked. Pillars pass through the insulating layers and the gate electrodes along a direction orthogonal to an upper surface of the substrate.
(6) A semiconductor device according to an exemplary embodiment of the present inventive concept includes a substrate having a main block region and a dummy block region which is continuous with the main block region. A plurality of main blocks are formed in the main block region on the substrate. A plurality of dummy blocks are formed in the dummy block region on the substrate. A plurality of main trenches are disposed between the main blocks. A plurality of dummy trenches are disposed between the dummy blocks. Main source lines are formed inside the main trenches, and dummy source lines are formed inside the dummy trenches. Each of the main blocks and the dummy blocks includes insulating layers and gate electrodes which are alternately and repeatedly stacked, Pillars are dimensioned and positioned to pass through the insulating layers and the gate electrodes along a direction orthogonal to an upper surface of the substrate. A lower end of an outermost dummy source line among the dummy source lines is formed at a level higher than levels of lower ends of the main source lines.

(7) A semiconductor device according to an exemplary embodiment of the present inventive concept includes a plurality of main blocks and a plurality of dummy blocks on a substrate. Main trenches are between the main blocks. Dummy trenches are between the dummy blocks. Main source lines are formed inside the main trenches. Dummy source lines are formed inside the dummy trenches. Each of the main blocks and the dummy blocks includes insulating layers and gate electrodes which are alternately and repeatedly stacked. Pillars are dimensioned and positioned to pass through the insulating layers and the gate electrodes along a direction orthogonal to an upper surface of the substrate. At least one dummy trench selected from among the dummy trenches has a width different from a width of each of the main trenches.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other objects, features and aspects of the present inventive concept will become more apparent to those of ordinary skill in the art by describing exemplary embodiments thereof in detail with reference to the accompanying drawings, in which:

(2) FIGS. **1** to **6** are cross-sectional views illustrating a semiconductor device according to an exemplary embodiment of the present inventive concept;

(3) FIGS. **7** and **8** are each partially enlarged views illustrating a part of FIG. **1**;

(4) FIG. **9** is a partial cross-sectional view of a semiconductor device according to an exemplary embodiment of the present inventive concept;

(5) FIG. **10** is a layout view of a semiconductor device according to an exemplary embodiment of the present inventive concept;

(6) FIGS. **11** to **18** are cross-sectional views taken along line I-I′ of FIG. **10** illustrating a method of forming a semiconductor device according to an exemplary embodiment of the present inventive concept; and

(7) FIG. **19** is a layout view of a semiconductor device according to an exemplary embodiment of the present inventive concept.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

(8) FIGS. **1** to **6** are cross-sectional views illustrating a semiconductor device according to an exemplary embodiment of the present inventive concept. FIGS. **7** and **8** are each partially enlarged views illustrating a part of FIG. **1**. A semiconductor device according to an exemplary embodiment of the present inventive concept may include a flash memory such as a vertical-NAND (VNAND) or a three dimensional NAND (3D-NAND) memory.

(9) Referring to FIG. **1**, a semiconductor device according to an exemplary embodiment of the present inventive concept may include a cell array region having a main block region MB and a dummy block region DB. The dummy block region DB may be continuous with the main block region MB. The dummy block region DB may be provided at an edge of the cell array region. As an example, a substrate **21** may include the dummy block region DB and the main block region MB.

(10) The semiconductor device may include insulating layers **25**, mold layers **27**, a first insulating interlayer **29**, pillars **41** and **42**, a second insulating interlayer **45**, trenches **131**, **132**, **141**, **142**, and **143**, impurity regions **47**, a gate insulating layer **53**, gate electrodes G1 to Gn, a spacer **55**, source lines **151**, **152**, **161**, **162**, and **163**, a third insulating interlayer **57**, first bit plugs **61**, sub-bit lines **63**, a fourth insulating interlayer **65**, second bit plugs **67**, and bit lines **69**, which are formed on a substrate **21** having the main block region MB and the dummy block region DB.

(11) The pillars **41** and **42** may include cell pillars **41** formed in the main block region MB and dummy pillars **42** formed in the dummy block region DB. Each of the pillars **41** and **42** may include a semiconductor pattern **31**, a channel structure **37**, and a conductive pad **39**. The channel

structure **37** may include an information storage pattern **33**, a channel pattern **34**, and a core pattern **35**.
(12) The trenches **131**, **132**, **141**, **142**, and **143** may include a first main trench **131**, a second main trench **132**, a first dummy trench **141**, a second dummy trench **142**, and a third dummy trench **143**. The trenches **131**, **132**, **141**, **142**, and **143** may be positioned in-parallel with each other. For example, an extending direction of each of the trenches **131**, **132**, **141**, **142**, and **143** may extend along a direction orthogonal to an upper surface of the substrate **21**, and the extending directions of the trenches **131**, **132**, **141**, **142**, and **143** may be substantially parallel with each other. In an exemplary embodiment of the present inventive concept, the trenches **131**, **132**, **141**, **142**, and **143** may be referred to as a word line cut (WLCUT) or a common source line (CSL) trench. A first main block MB1, a second main block MB2, a third main block MB3, a first dummy block DB1, a second dummy block DB2, and a third dummy block DB3 may be defined by the trenches **131**, **132**, **141**, **142**, and **143**.
(13) The first main block MB1, the second main block MB2, and the third main block MB3 may be defined in the main block region MB. The first dummy block DB1, the second dummy block DB2, and the third dummy block DB3 may be defined in the dummy block region DB. In an exemplary embodiment of the present inventive concept, the second dummy block DB2 and the third dummy block DB3 may be selectively omitted. In an exemplary embodiment of the present inventive concept, the dummy block region DB may include 4 to 8 dummy blocks, but exemplary embodiments of the present inventive concept are not limited thereto.
(14) Each of the first main block MB1, the second main block MB2, the third main block MB3, the first dummy block DB1, the second dummy block DB2, and the third dummy block DB3 may include a stack structure in which the insulating layers **25** and the gate electrodes G1 to Gn are alternately and repeatedly stacked. Each of the first main block MB1, the second main block MB2, and the third main block MB3 may include the cell pillars **41** perpendicularly passing through the insulating layers **25** and the gate electrodes G1 to Gn (e.g., along the direction orthogonal to the upper surface of the substrate **21**). Each of the first dummy block DB1, the second dummy block DB2, and the third dummy block DB3 may include the dummy pillars **42**. The mold layers **27** may partially remain in the first dummy block DB1 and the second dummy block DB2. The mold layers **27** may be disposed between the insulating layers **25**. Each of the mold layers **27** may be formed at a level substantially the same as a level of each of the gate electrodes G1 to Gn. The mold layers **27** may be in direct contact with side surfaces of the gate electrodes G1 to Gn. As an example, each of the mold layers **27** may be disposed on (e.g., may be in direct contact with) a portion of an upper surface of a respective insulating layer **25**. For example, each of the mold layers **27** may be disposed on (e.g., may be in direct contact with) a portion of an upper surface of a respective insulating layer **25** on which a respective gate electrode (e.g., one of gate electrodes G1 to Gn) is not disposed. Thus, a respective gate electrode may be disposed on a first portion of a respective insulating layer **25**, while a mold layer **27** is disposed on a second portion of the respective insulating layer **25**.
(15) A first gate electrode G1 may correspond to a ground selection line GSL. Each of a second gate electrode G2 to an (n−2).sup.th gate electrode Gn−2 may correspond to a control gate line. Each of an (n−1).sup.th gate electrode Gn−1 and an n.sup.th gate electrode Gn may correspond to a string selection line (SSL) or a drain selection line (DSL).
(16) The first dummy block DB1 may be defined in an outermost region of the dummy block region DB (e.g., a region of the dummy block region relatively furthest away from the main block region MB). The first dummy trench **141** may be formed between the first dummy block DB1 and the second dummy block DB2. The first dummy trench **141** may be formed as an outermost trench among the trenches **131**, **132**, **141**, **142**, and **143**. The first main trench **131**, the second main trench **132**, the second dummy trench **142**, and the third dummy trench **143** may perpendicularly pass completely through the insulating layers **25** and the gate electrodes G1 to Gn (e.g., along the

direction orthogonal to the upper surface of the substrate **21**). Lower ends of the first main trench **131**, the second main trench **132**, the second dummy trench **142**, and the third dummy trench **143** may be formed at levels lower than a level of an upper end of the substrate **21**.

(17) A lower end of the first dummy trench **141** may be formed at a level higher than the levels of the lower ends of the first main trench **131**, the second main trench **132**, the second dummy trench **142**, and the third dummy trench **143**. In an exemplary embodiment of the present inventive concept, the lower end of the first dummy trench **141** may be formed at a level higher than a level of a lowest layer among the mold layers **27**. Lateral widths of the first main trench **131**, the second main trench **132**, the second dummy trench **142**, and the third dummy trench **143** may be substantially the same as each other. For example, the lateral widths may be the same as each other along a direction parallel to the upper surface of the substrate **21**. A lateral width of the first dummy trench **141** may be smaller than the lateral width of each of the first main trench **131**, the second main trench **132**, the second dummy trench **142**, and the third dummy trench **143**.

(18) In an exemplary embodiment of the present inventive concept, the lower end of the first dummy trench **141** may be formed at a level higher than a level of the first gate electrode G1. The lower end of the first dummy trench **141** may be formed between the first gate electrode G1 and the second gate electrode G2. The lower end of the first dummy trench **141** may be formed between the lowest layer among the mold layers **27** and the second gate electrode G2.

(19) The impurity regions **47** may be in a position corresponding to a common source region. The source lines **151**, **152**, **161**, **162**, and **163** may include a first main source line **151**, a second main source line **152**, a first dummy source line **161**, a second dummy source line **162**, and a third dummy source line **163**.

(20) The first main source line **151**, the second main source line **152**, the second dummy source line **162**, and the third dummy source line **163** may each be in direct contact with a respective one of the impurity regions **47**. A lower end of the first dummy source line **161** may be in direct contact with a second lowest layer among the insulating layers **25**. For example, the lower end of the first dummy source line **161** may be positioned between an upper surface and a lower surface of the second lowest layer among the insulating layers **25**.

(21) Lower ends of the first main source line **151**, the second main source line **152**, the second dummy source line **162**, and the third dummy source line **163** may be formed at levels lower than the level of the upper end of the substrate **21**. The lower end of the first dummy source line **161** may be formed at a level higher than the levels of the lower ends of the first main source line **151**, the second main source line **152**, the second dummy source line **162**, and the third dummy source line **163**. In an exemplary embodiment of the present inventive concept, the lower end of the first dummy source line **161** may be formed at a level higher than the lowest layer among the mold layers **27**. Lateral widths of the first main source line **151**, the second main source line **152**, the second dummy source line **162**, and the third dummy source line **163** may be substantially the same as each other. A lateral width of the first dummy source line **161** may be smaller than the lateral width of each of the first main source line **151**, the second main source line **152**, the second dummy source line **162**, and the third dummy source line **163**.

(22) In an exemplary embodiment of the present inventive concept, the lower end of the first dummy source line **161** may be formed at a level higher than the level of the first gate electrode G1. The lower end of the first dummy source line **161** may be formed between the first gate electrode G1 and the second gate electrode G2. The lower end of the first dummy source line **161** may be formed between the lowest layer among the mold layers **27** and the second gate electrode G2.

(23) According to another embodiment of the present inventive concept (see, e.g., FIG. **1**), a semiconductor device may include the substrate **21** including the main block region MB and the dummy block region DB. At least two main block pillars (e.g., **41**) may be positioned in the main block region MB. At least two dummy pillars (e.g., **42**) may be positioned in the dummy block

region DB. At least two main block trenches (e.g., **131** and **132**) may be positioned in the main block region MB. The at least two main block pillars may separate the at least two main block trenches from each other. A main source line (e.g., **151** and **152**) may be positioned in each of the at least two main block trenches. At least two dummy block trenches **141**, **142** and/or **143**) may be positioned in the dummy block region DB. A dummy source line (e.g., **161**, **162** or **163**) may be positioned in each of the at least two dummy block trenches. Gate electrodes (e.g., Gn to G1) and insulating layers (e.g., **25**) may be alternately and repeatedly stacked above the substrate **21**. An extending direction of each of the gate electrodes and insulating layers may be perpendicular to an extending direction of the at least two main block pillars and the at least two dummy pillars. A lowermost level of one of the at least two dummy block trenches (e.g., **141**) may be spaced further apart from an upper surface of the substrate **21** than a lowermost level of each of the at least two main block trenches (e.g., **131** and **132**).
(24) Referring to FIG. **2**, a lateral width of a first dummy trench **141** may be substantially the same as a lateral width of each of a first main trench **131**, a second main trench **132**, a second dummy trench **142**, and a third dummy trench **143**. A lower end of the first dummy trench **141** may be formed at a level higher than levels of lower ends of the first main trench **131**, the second main trench **132**, the second dummy trench **142**, and the third dummy trench **143**. In an exemplary embodiment of the present inventive concept, the lower end of the first dummy trench **141** may be formed at a level higher than a level of a lowest layer among mold layers **27**.
(25) A lateral width of a first dummy source line **161** may be substantially the same as a lateral width of each of a first main source line **151**, a second main source line **152**, a second dummy source line **162**, and a third dummy source line **163**. The lower end of the first dummy source line **161** may be formed at a level higher than levels of lower ends of the first main source line **151**, the second main source line **152**, the second dummy source line **162**, and the third dummy source line **163**. In an exemplary embodiment of the present inventive concept, the lower end of the first dummy source line **161** may be formed at a level higher than the level of the lowest layer among the mold layers **27**.
(26) Referring to FIG. **3**, a lower end of a second dummy trench **142** may be formed at a level higher than levels of lower ends of a first main trench **131**, a second main trench **132**, and a third dummy trench **143**. The lower end of the second dummy trench **142** may be formed at a level lower than a level of a lower end of a first dummy trench **141**. In an exemplary embodiment of the present inventive concept, the second dummy trench **142** may be positioned and dimensioned to pass through a first gate electrode G1 (e.g., along the direction orthogonal to the upper surface of the substrate **21**), and the lower end of the second dummy trench **142** may be formed at a level higher than a level of an upper end of a substrate **21**.
(27) A lower end of a second dummy source line **162** may be formed at a level higher than levels of lower ends of a first main source line **151**, a second main source line **152**, and a third dummy source line **163**. The lower end of the second dummy source line **162** may be formed at a level lower than a level of lower end of a first dummy source line **161**. In an exemplary embodiment of the present inventive concept, the lower end of the second dummy source line **162** may be formed at a level higher than the level of the upper end of the substrate **21**.
(28) Referring to FIG. **4**, a lower end of a second dummy trench **142** may be formed at a level higher than levels of lower ends of a first main trench **131**, a second main trench **132**, and a third dummy trench **143**. The lower end of the second dummy trench **142** may be formed at a level lower than a level of a lower end of a first dummy trench **141**. In an exemplary embodiment of the present inventive concept, the lower ends of the second dummy trench **142** and the first dummy trench **141** may be formed at levels lower than a level of an upper end of a substrate **21**.
(29) A lower end of a second dummy source line **162** may be formed at a level higher than levels of lower ends of a first main source line **151**, a second main source line **152**, and a third dummy source line **163**. The lower end of the second dummy source line **162** may be formed at a level

lower than a level of a lower end of a first dummy source line **161**. In an exemplary embodiment of the present inventive concept, the lower ends of the second dummy source line **162** and the first dummy source line **161** may be formed at levels lower than the level of the upper end of the substrate **21**.

(30) Referring to FIG. **5**, the semiconductor device may include lower insulating layers **25**, lower mold layers **27**, first insulating interlayer **29**, lower pillars **41** and **42**, upper insulating layers **225**, upper mold layers **227**, upper pillars **241** and **242**, second insulating interlayer **45**, trenches **131**, **132**, **141**, **142**, and **143**, impurity regions **47**, gate insulating layer **53**, gate electrodes G1 to Gn and Gm−4 to Gm, spacer SS, source lines **151**, **152**, **161**, **162**, and **163**, third insulating interlayer **57**, first bit plugs **61**, sub-bit lines **63**, fourth insulating interlayer **65**, a fifth insulating interlayer **229**, second bit plugs **67**, and bit lines **69**, which are formed on the substrate **21**.

(31) The lower insulating layers **25**, the lower mold layers **27**, the first insulating interlayer **29**, the gate electrodes G1 to Gn, and the lower pillars **41** and **42** may be included in a lower stack structure. The upper insulating layers **225**, the upper mold layers **227**, the gate electrodes Gm−4 to Gm, the upper pillars **241** and **242**, and the fifth insulating interlayer **229** may be included in an upper stack structure. The semiconductor device according to an exemplary embodiment of the present inventive concept may have a dual-stack structure. In an exemplary embodiment of the present inventive concept, the semiconductor device may have a multi-stack structure.

(32) The lower pillars **41** and **42** may include lower cell pillars **41** and lower dummy pillars **42**. Each of the lower pillars **41** and **42** may include a semiconductor pattern **31**, a lower channel structure **37**, and a lower conductive pad **39**. The lower channel structure **37** may include a lower information storage pattern **33**, a lower channel pattern **34**, and a lower core pattern **35**. The upper pillars **241** and **242** may include upper cell pillars **241** formed on the lower cell pillars **41** and upper dummy pillars **242** formed on the lower dummy pillars **42**. Each of the upper pillars **241** and **242** may include an upper channel structure **237** and an upper conductive pad **239**. Upper conductive pads **239** may be disposed on top surfaces of each of the upper pillars **241** and **242**. The upper channel structure **237** may include an upper information storage pattern **233**, an upper channel pattern **234**, and an upper core pattern **235**.

(33) The upper insulating layers **225**, the upper mold layers **227**, the fifth insulating interlayer **229**, the gate electrodes Gm−4 to Gm, the upper channel structure **237**, and the upper conductive pad **239** may include materials which are substantially the same as materials of the lower insulating layers **25**, the lower mold layers **27**, the first insulating interlayer **29**, the gate electrodes G1 to Gn, the lower channel structure **37**, and the lower conductive pad **39**, respectively, and may be respectively formed using a similar method as for forming the lower insulating layers **25**, the lower mold layers **27**, the first insulating interlayer **29**, the gate electrodes G1 to Gn, the lower channel structure **37**, and the lower conductive pad **39**.

(34) Referring to FIG. **6**, the semiconductor device may include lower insulating layers **25**, lower mold layers **27**, first insulating interlayer **29**, upper insulating layers **225**, upper mold layers **227**, pillars **41** and **42**, second insulating interlayer **45**, trenches **131**, **132**, **141**, **142**, and **143**, impurity regions **47**, gate insulating layer **53**, gate electrodes G1 to Gn and Gm−4 to Gn, spacer **55**, source lines **151**, **152**, **161**, **162**, and **163**, third insulating interlayer **57**, first bit plugs **61**, sub-bit lines **63**, fourth insulating interlayer **65**, fifth insulating interlayer **229**, second bit plugs **67**, and bit lines **69**, which are formed on the substrate **21**.

(35) The lower insulating layers **25**, the lower mold layers **27**, the first insulating interlayer **29**, and the gate electrodes G1 to Gn may be included in a lower stack structure. The upper insulating layers **225**, the upper mold layers **227**, the gate electrodes Gm−4 to Gm, and the fifth insulating interlayer **229** may be included in an upper stack structure. The semiconductor device according to an exemplary embodiment of the present inventive concept may have a dual-stack structure. In an exemplary embodiment of the present inventive concept, the semiconductor device may have a multi-stack structure.

(36) The pillars **41** and **42** may include cell pillars **41** and dummy pillars **42**. Each of the pillars **41** and **42** may include semiconductor pattern **31**, channel structure **37**, and conductive pad **239**. The channel structure **37** may include information storage pattern **33**, channel pattern **34**, and core pattern **35**. Each of the pillars **41** and **42** may pass through the upper stack structure and the lower stack structure.
(37) According to the an exemplary embodiment of the present inventive concept, the main trenches **131** and **132** defining the main blocks MB1, MB2, and MB3 and the dummy trenches **141**, **142**, and **143** defining the dummy blocks DB1, DB2, and DB3 may be formed. The main source lines **151** and **152** may be respectively formed inside the main trenches **131** and **132**, and the dummy source lines **161**, **162**, and **163** may be respectively formed inside the dummy trenches **141**, **142**, and **143**. The lower end of the first dummy trench **141** may be formed at a level higher than the levels of the lower ends of the main trenches **131** and **132**. The lower end of the first dummy source line **161** may be formed at a level higher than the levels of the lower ends of the main source lines **151** and **152** . . . . Due to a configuration of the first dummy trench **141** and the first dummy source line **161**, the number of defects occurring at the edge of the cell array region (e.g., at the outermost edge of the dummy block region DB) can be significantly reduced. Thus, sizes and the number of the dummy blocks DB1, DB2, and DB3 can be minimized.
(38) Referring to FIG. **7**, the information storage pattern **33** may include a tunnel insulating layer **71**, a charge storage layer **72** which surrounds an outer side of the tunnel insulating layer **71**, and a first blocking layer **73** which surrounds an outer side of the charge storage layer **72**. A second blocking layer **75** may cover upper surfaces and lower surfaces of gate electrodes G1 to Gn and may be disposed between the gate electrodes G1 to Gn and the first blocking layer **73**.
(39) Referring to FIG. **8**, the information storage pattern **33** may include the tunnel insulating layer **71**, the charge storage layer **72** which surrounds an outer side of the tunnel insulating layer **71**, and the first blocking layer **73** which surrounds an outer side of the charge storage layer **72**. The first blocking layer **73** may be in direct contact with gate electrodes G1 to Gn. The second blocking layer **75** (see FIG. **7**) may be omitted and thus the first blocking layer **73** may be in direct contact with gate electrodes G1 to Gn when the second blocking layer **75** is omitted.
(40) FIG. **9** is a partial cross-sectional view of a semiconductor device according to an exemplary embodiment of the present inventive concept.
(41) Referring to FIG. **9**, bottoms of trenches **131**, **132**, **141**, **142**, and **143** may be formed to be rounded and pointed. In an exemplary embodiment of the present inventive concept, lower regions of the trenches **131**, **132**, **141**, **142**, and **143** may be formed to be tapered along a downward direction (e.g., along the direction orthogonal to the upper surface of the substrate **21**). Lower ends of source lines **151**, **152**, **161**, **162**, and **163** may be formed to be rounded and pointed. In an exemplary embodiment of the present inventive concept, lower regions of the source lines **151**, **152**, **161**, **162**, and **163** may be formed to be tapered along a downward direction (e.g., along the direction orthogonal to the upper surface of the substrate **21**).
(42) FIG. **10** is a layout view of a semiconductor device according to an exemplary embodiment of the present inventive concept. FIGS. **11** to **18** are cross-sectional views taken along line I-P of FIG. **10** illustrating a method of forming a semiconductor device according to an exemplary embodiment of the present inventive concept. In an exemplary embodiment of the present inventive concept, FIG. **10** may illustrate a part of a cell array region of a VNAND memory.
(43) Referring to FIGS. **10** and **11**, insulating layers **25** and mold layers **27** may be formed on the substrate **21** having the main block region MB and the dummy block region DB. The insulating layers **25** and the mold layers **27** may be alternately and repeatedly stacked.
(44) The substrate **21** may include a semiconductor substrate such as a silicon wafer or a silicon-on-insulator (SOI) wafer. For example, the substrate **21** may be a single crystalline silicon wafer including P-type impurities such as boron (B).
(45) The mold layers **27** may include a material having etch selectivity with respect to the

insulating layers **25**. For example, the insulating layers **25** may include silicon oxide, and the mold layers **27** may include silicon nitride. In an exemplary embodiment of the present inventive concept, the insulating layers **25** and the mold layers **27** may be formed in the same chamber as each other using an in-situ process. As an example, the insulating layers **25** and the mold layers **27** may be formed using various types of a chemical vapor deposition (CVD) method or an atomic layer deposition (ALD) method.
(46) Referring to FIGS. **10** and **12**, in an edge of the dummy block region DB, edges of the insulating layers **25** and the mold layers **27** may be patterned and have a stepped structure. A first insulating interlayer **29** may be formed on the substrate **21**. The first insulating interlayer **29** may cover the edges of the insulating layers **25** and the mold layers **27**, which are formed to have the stepped structure. The edges of the insulating layers **25** and the mold layers **27** may face along a direction parallel to an upper surface of the substrate **21**. The first insulating interlayer **29** may include an insulating material such as silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof.
(47) Referring to FIGS. **10** and **13**, pillars **41** and **42** may be formed to pass through the insulating layers **25** and the mold layers **27** (e.g., along the direction orthogonal to the upper surface of the substrate **21**). The pillars **41** and **42** may include cell pillars **41** formed in the main block region MB and dummy pillars **42** formed in the dummy block region DB. Each of the pillars **41** and **42** may include the semiconductor pattern **31**, the channel structure **37**, and the conductive pad **39**. The channel structure **37** may include the information storage pattern **33**, the channel pattern **34**, and the core pattern **35**.
(48) The semiconductor pattern **31** may be in direct contact with the substrate **21**. For example, the semiconductor pattern **31** may be in direct contact with an upper portion of the substrate **21**. A lower surface of the semiconductor pattern **31** may be positioned below an upper surface of the substrate **21**. The semiconductor pattern **31** may be formed using a selective epitaxial growth (SEG) process. In an exemplary embodiment of the present inventive concept, the semiconductor pattern **31** may include single crystalline silicon having P-type impurities. The channel structure **37** may be formed on the semiconductor pattern **31**. A process of forming a plurality of thin films and an etch-back process may be applied as a process of forming the channel structure **37**.
(49) The core pattern **35** may include an insulator including silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof. In an exemplary embodiment of the present inventive concept, the core pattern **35** may include polysilicon. The channel pattern **34** may surround a side surface and a lower portion of the core pattern **35**. The channel pattern **34** may include a semiconductor layer such as a polysilicon layer. The channel pattern **34** may be in direct contact with the semiconductor pattern **31**. The information storage pattern **33** may be formed to surround an outer side of the channel pattern **34**.
(50) In an exemplary embodiment of the present inventive concept, (see, e.g., FIGS. **7** and **8**), the information storage pattern **33** may include the tunnel insulating layer **71**, the charge storage layer **72** which surrounds an outer side of the tunnel insulating layer **71**, and the first blocking layer **73** which surrounds an outer side of the charge storage layer **72**. The information storage pattern **33** may include a plurality of insulating layers including silicon oxide, silicon nitride, silicon oxynitride, a high-K dielectric, or a combination thereof. In an exemplary embodiment of the present inventive concept, the tunnel insulating layer **71** may include silicon oxide, the charge storage layer **72** may include silicon nitride, and the first blocking layer **73** may include aluminum oxide (AlO).
(51) The conductive pad **39** may be formed on the channel structure **37**. The conductive pad **39** may be formed using a process of forming a thin film and a planarization process. The planarization process may include a chemical mechanical polishing (CMP) process, an etch-back process, or a combination thereof. The conductive pad **39** may be in direct contact with the channel structure **37**. The conductive pad **39** may include a conductive material such as polysilicon, a metal,

a metal silicide, a metal oxide, a metal nitride, conductive carbon, or a combination thereof.

(52) Referring to FIGS. **10** and **14**, the second insulating interlayer **45** may be formed to cover the pillars **41** and **42**. The second insulating interlayer **45** may include an insulating layer including silicon oxide, silicon nitride, silicon oxynitride, a low-K dielectric, or a combination thereof. Trenches **131**, **132**, **141**, **142**, and **143** may be formed to pass through the second insulating interlayer **45**, the insulating layers **25**, and the mold layers **27** (e.g., along the direction orthogonal to the upper surface of the substrate **21**). The trenches **131**, **132**, **141**, **142**, and **143** may include a first main trench **131**, a second main trench **132**, a first dummy trench **141**, a second dummy trench **142**, and a third dummy trench **143**. A first main block MB1, a second main block MB2, a third main block MB3, a first dummy block DB1, a second dummy block DB2, and a third dummy block DB3 may be defined by the trenches **131**, **132**, **141**, **142**, and **143**.

(53) A patterning process may be applied as a process of forming the trenches **131**, **132**, **141**, **142**, and **143**. A hard mask pattern may be formed on the second insulating interlayer **45**.

(54) The first dummy block DB1 may be defined in an outermost region of the dummy block region DB (e.g., a region of the dummy block region DB relatively furthest away from the main block region MB). The first dummy trench **141** may be formed between the first dummy block DB1 and the second dummy block DB2. The first dummy trench **141** may be formed as an outermost trench among the trenches **131**, **132**, **141**, **142**, and **143**. The first main trench **131**, the second main trench **132**, the second dummy trench **142**, and the third dummy trench **143** may perpendicularly pass completely through the insulating layers **25** and the mold layers **27** (e.g., along the direction orthogonal to the upper surface of the substrate **21**). Lower ends of the first main trench **131**, the second main trench **132**, the second dummy trench **142**, and the third dummy trench **143** may be formed at levels lower than a level of an upper end of the substrate **21**.

(55) A lower end of the first dummy trench **141** may be formed at a level higher than the levels of the lower ends of the first main trench **131**, the second main trench **132**, the second dummy trench **142**, and the third dummy trench **143**. In an exemplary embodiment of the present inventive concept, the lower end of the first dummy trench **141** may be formed at a level higher than a level of a lowest layer among the mold layers **27**. A second layer among the insulating layers **25** may be exposed at a bottom of the first dummy trench **141**. Lateral widths of the first main trench **131**, the second main trench **132**, the second dummy trench **142**, and the third dummy trench **143** may be substantially the same as each other. A lateral width of the first dummy trench **141** may be smaller than the lateral width of each of the first main trench **131**, the second main trench **132**, the second dummy trench **142**, and the third dummy trench **143**.

(56) Impurity regions **47** may be formed in the substrate **21** at which lower portions of the first main trench **131**, the second main trench **132**, the second dummy trench **142**, and the third dummy trench **143** are exposed. The impurity regions **47** may be formed using an ion implanting process. In an exemplary embodiment of the present inventive concept, the impurity regions **47** may include N-type impurities such as phosphorus (P) or arsenic (As). The impurity regions **47** may correspond to a common source region.

(57) Referring to FIGS. **10** and **15**, openings **49** which communicate with (e.g., are connected with) the trenches **131**, **132**, **141**, **142**, and **143** may be formed by removing the mold layers **27**. An isotropic etching process may be used as a process of removing the mold layers **27**. The mold layers **27** may partially remain in the first dummy block DB1 and the second dummy block DB2.

(58) Referring to FIGS. **10** and **16**, a gate insulating layer **53** and gate electrodes G1 to Gn may be formed.

(59) The gate insulating layer **53** may be formed on a side surface of the semiconductor pattern **31** exposed inside the openings **49**. In an exemplary embodiment of the present inventive concept, the gate insulating layer **53** may include a thermal oxide layer. A process of forming a thin film and an etching process may be applied as a process of forming the gate electrodes G1 to Gn. The etching process may include an anisotropic etching process, an isotropic etching process, or a combination

thereof. The gate electrodes G1 to Gn may be formed inside the openings **49**. The gate electrodes G1 to Gn may include a conductive material such as a metal, a metal silicide, a metal oxide, a metal nitride, polysilicon, conductive carbon, or a combination thereof. For example, the gate electrodes G1 to Gn may include Ti, TiN, Ta, TaN, W, WN, Ru, Pt, or a combination thereof.
(60) In an exemplary embodiment of the present inventive concept, before the gate electrodes G1 to Gn are formed, the second blocking layer **75** may be formed (see, e.g., FIG. **7**). The second blocking layer **75** may cover upper surfaces and lower surfaces of the gate electrodes G1 to Gn and may be disposed between the gate electrodes G1 to Gn and the first blocking layer **73**. The second blocking layer **75** may include an insulating layer including silicon oxide, silicon nitride, silicon oxynitride, a high-K dielectric, or a combination thereof.
(61) Referring to FIGS. **10** and **17**, the spacer **55** and source lines **151**, **152**, **161**, **162**, and **163** may be formed inside the trenches **131**, **132**, **141**, **142**, and **143**, The source lines **151**, **152**, **161**, **162**, and **163** may include a first main source line **151**, a second main source line **152**, a first dummy source line **161**, a second dummy source line **162**, and a third dummy source line **163**.
(62) The spacer **55** may be formed using a process of forming a thin film and an anisotropic etching process. The spacer **55** may cover side walls of the trenches **131**, **132**, **141**, **142**, and **143**. The spacer **55** may include an insulating layer including silicon oxide, silicon nitride, silicon oxynitride, a high-K dielectric, a low-K dielectric, or a combination thereof. While a process of forming the spacer **55** is performed, bottoms of the trenches **131**, **132**, **141**, **142**, and **143** may be recessed downward. The source lines **151**, **152**, **161**, **162**, and **163** may substantially fill the trenches **131**, **132**, **141**, **142**, and **143**.
(63) A process of forming a thin film and a planarization process may be applied as a process of forming the source lines **151**, **152**, **161**, **162**, and **163**. The source lines **151**, **152**, **161**, **162**, and **163** may include a conductive material such as a metal, a metal silicide, a metal oxide, a metal nitride, polysilicon, conductive carbon, or a combination thereof. For example, the source lines **151**, **152**, **161**, **162**, and **163** may include a W layer and a TiN layer which surrounds a side surface and a lower end of the W layer. The spacer **55** may be disposed between the source lines **151**, **152**, **161**, **162**, and **163** and the gate electrodes G1 to Gn. The first main source line **151**, the second main source line **152**, the second dummy source line **162**, and the third dummy source line **163** may be in direct contact with the impurity region **47**. A lower end of the first dummy source line **161** may be in direct contact with a second layer among the insulating layers **25**. For example, the lower end of the first dummy source line **161** may be positioned between upper and lower surfaces of a second layer among the insulating layers **25**.
(64) Lower ends of the first main source line **151**, the second main source line **152**, the second dummy source line **162**, and the third dummy source line **163** may be formed at levels lower than a level of an upper end of the substrate **21**. The lower end of the first dummy source line **161** may be formed at a level higher than the levels of the lower ends of the first main source line **151**, the second main source line **152**, the second dummy source line **162**, and the third dummy source line **163**. In an exemplary embodiment of the present inventive concept, the lower end of the first dummy source line **161** may be formed at a level higher than a level of a lowest layer among the mold layers **27**. Lateral widths of the first main source line **151**, the second main source line **152**, the second dummy source line **162**, and the third dummy source line **163** may be substantially the same as each other. A lateral width of the first dummy source line **161** may be smaller than the lateral width of each of the first main source line **151**, the second main source line **152**, the second dummy source line **162**, and the third dummy source line **163**.
(65) In an exemplary embodiment of the present inventive concept, each of the source lines **151**, **152**, **161**, **162**, and **163** may include a plurality of source plugs.
(66) Referring to FIGS. **10** and **18**, the third insulating interlayer **57** may be formed on the source lines **151**, **152**, **161**, **162**, and **163** and the second insulating interlayer **45**. First bit plugs **61** may be formed to pass through the third insulating interlayer **57** and the second insulating interlayer **45**

(e.g., along the direction orthogonal to the upper surface of the substrate **21**) to be in direct contact with the cell pillars **41**. Sub-bit lines **63** may be formed on the third insulating interlayer **57** to be in direct contact with the first bit plugs **61**.

(67) The third insulating interlayer **57** may include an insulating layer including silicon oxide, silicon nitride, silicon oxynitride, a low-K dielectric, or a combination thereof. The first bit plugs **61** and the sub-bit lines **63** may include a conductive material such as a metal, a metal silicide, a metal oxide, a metal nitride, polysilicon, conductive carbon, or a combination thereof.

(68) Referring back to FIGS. **1** and **10**, the fourth insulating interlayer **65** may be formed to cover the sub-bit lines **63** on the third insulating interlayer **57**. Second bit plugs **67** may be formed to pass through the fourth insulating interlayer **65** to be in direct contact with the sub-bit lines **63**. Bit lines **69** may be formed on the fourth insulating interlayer **65** to be in direct contact with the second bit plugs **67**.

(69) The fourth insulating interlayer **65** may include an insulating layer including silicon oxide, silicon nitride, silicon oxynitride, a low-K dielectric, or a combination thereof. The second bit plugs **67** and the bit lines **69** may include conductive material such as a metal, a metal silicide, a metal oxide, a metal nitride, polysilicon, conductive carbon, or a combination thereof.

(70) FIG. **19** is a layout view of a semiconductor device according to an exemplary embodiment of the present inventive concept.

(71) Referring to FIG. **19**, a semiconductor device according to an exemplary embodiment of the present inventive concept may include a cell array region including the main block region MB and the dummy block region DB. The semiconductor device may include first to ninth channel holes H1, H2, H3, H4, H5, H6, H7, H8, and H9, first to ninth cell pillars **41**A, **41**B, **41**C, **41**D, **41**E, **41**F, **41**G, **41**H, and **41**I, the first main trench **131**, the second main trench **132**, the first dummy trench **141**, the first main source line **151**, the second main source line **152**, the first dummy source line **161**, a sub-trench **435**, an insulating pattern **455**, and bit lines **69**, which are formed on the substrate **21** including the main block region MB and the dummy block region DB.

(72) A second main block MB2 may be defined between the first main trench **131** and the second main trench **132**. The first to ninth cell pillars **41**A, **41**B, **41**C, **41**D, **41**E, **41**F, **41**G, **41**H, and **41**I may be formed inside the first to ninth channel holes H1, H2, H3, H4, H5, H6, H7, H8, and H9. The second main block MB2 may include the first to fifth channel holes H1, H2, H3, H4, and H5 arranged in a first row direction, the sixth to ninth channel holes H6, H7, H8, and H9 arranged in a second row direction, and the insulating pattern **455** formed inside the sub-trench **435**. The first to ninth cell pillars **41**A, **41**B, **41**C, **41**D, **41**E, **41**F, **41**G, **41**H, and **41**I may include a configuration similar to the configuration of the cell pillars **41** described with reference to FIGS. **1** to **6**. The insulating pattern **455** may include silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof.

(73) The first channel hole H1 may be adjacent to the first main trench **131**, the fifth channel bole H5 may be adjacent to the second main trench **132**, the third channel hole H3 may be arranged between the first channel hole H1 and the fifth channel hole H5, the second channel hole H2 may be arranged between the first channel hole H1 and the third channel hole H3, and the fourth channel hole H4 may be arranged between the third channel hole H3 and the fifth channel hole H5. The sixth channel hole H6 may be adjacent to the first main trench **131**, the ninth channel hole H9 may be adjacent to the second main trench **132**, the seventh channel hole H7 may be arranged between the sixth channel hole H6 and the ninth channel hole H9, and the eighth channel bole H8 may be arranged between the seventh channel hole H7 and the ninth channel hole H9. The sixth channel hole H6 may be arranged between the first channel hole H1 and the second channel hole H2, the seventh channel bole H7 may be arranged between the second channel hole H2 and the third channel hole H3, the eighth channel hole H8 may be arranged between the third channel hole H3 and the fourth channel hole H4, and the ninth channel hole H9 may be arranged between the fourth channel hole H4 and the fifth channel hole H5.

(74) The first main trench **131** may be parallel to the second main trench **132**. The sub-trench **435** may be disposed between the first main trench **131** and the second main trench **132**. The sub-trench **435** may cross the third channel hole H3 and may extend between the seventh channel hole H7 and the eighth channel hole H8. The second main block MB2 may be divided into a first sub-block MB21 and a second sub-block MB22 by the sub-trench **435**. The first sub-block MB21 may include the first channel hole H1, the second channel hole H2, the sixth channel hole H6, and the seventh channel hole H7. The second sub-block MB22 may include the fourth channel hole H4, the fifth channel hole H5, the eighth channel hole H8, and the ninth channel hole H9.
(75) According to an exemplary embodiment of the present inventive concept, main trenches defining main blocks and dummy trenches defining dummy blocks may be provided. Main source lines may be formed inside the main trenches, and dummy source lines may be formed inside the dummy trenches. A lower end of an outermost dummy trench among the dummy trenches may be formed at a level higher than levels of lower ends of the main trenches. A lower end of an outermost dummy source line among the dummy source lines may be formed at a level higher than levels of lower ends of the main source lines. The number of defects occurring at edges of a cell array region can be significantly reduced by the configurations of the outermost dummy trench and the outermost dummy source line. Sizes and the number of the dummy blocks can be minimized. Thus, a semiconductor device having a relatively high degree of integration with a relatively low occurrence of defects may be realized.
(76) While the inventive concept has been particularly shown and described with reference to exemplary embodiments thereof, it will be understood by those of ordinary skill in the art that various changes in form and detail may be made therein without departing from the spirit and scope of the inventive concept.

Claims

1. A semiconductor device comprising: a substrate having a first surface and a second surface facing each other; a first main block and a second main block on the first surface of the substrate; a first dummy block and a second dummy block on the first surface of the substrate; a main trench between the first main block and the second main block; and a dummy trench between a portion of the first dummy block and a portion of the second dummy block, wherein each of the first and second main blocks includes: main insulating layers and main conductive material layers, which are alternately and repeatedly stacked, and cell pillars penetrating through the main insulating layers and the main conductive material layers in a vertical direction perpendicular to the first surface of the substrate, wherein each of the first and second dummy blocks includes: dummy insulating layers and dummy conductive material layers, which are alternately and repeatedly stacked; a lower mold layer between the substrate and a lowermost dummy conductive material layer of the dummy conductive material layers, wherein a material of the lower mold layer is different from a material of each of the main conductive material layers and the dummy conductive material layers; and dummy pillars penetrating through the dummy insulating layers, the dummy conductive material layers, and the lower mold layer in the vertical direction, wherein the first and second main blocks are adjacent to each other in a first direction parallel to the first surface of the substrate, wherein the first and second dummy blocks are adjacent to each other in the first direction, and wherein the lower mold layer of each of the first and second dummy blocks is substantially same level as a lowermost main conductive material layer of the main conductive material layers.
2. The semiconductor device of claim 1, wherein each of the main trench and the dummy trench extends in a second direction parallel to the first surface of the substrate, and wherein the second direction is perpendicular to the first direction.
3. The semiconductor device of claim 2, wherein a width of the main trench in the first direction is

greater than a width of the dummy trench in the first direction.
4. The semiconductor device of claim 1, wherein the lower mold layer of the first dummy block and the lower mold layer of the second dummy block extend between a lower surface of the dummy trench and the first surface of the substrate.
5. The semiconductor device of claim 1, wherein a lower end of the dummy trench is spaced apart from the substrate and is at a higher level than the first surface of the substrate, and wherein the main trench extends in the substrate.
6. The semiconductor device of claim 1, wherein a distance between the dummy trench and the second surface of the substrate is greater than a distance between the main trench and the second surface of the substrate.
7. The semiconductor device of claim 1, wherein an uppermost main conductive material layer of the main conductive material layers is substantially at the same level as an uppermost dummy conductive material layer of the dummy conductive material layers, wherein the lowermost dummy conductive material layer of the dummy conductive material layers is at a higher level than the lowermost main conductive material layer, and wherein the lower mold layer is between the lowermost dummy conductive material layer and the substrate.
8. The semiconductor device of claim 1, wherein the material of the lower mold layer is an insulating material having an etch selectivity with the dummy insulating layers.
9. The semiconductor device of claim 1, further comprising: a bit line on the first main block, the second main block, the first dummy block, and the second dummy block; and bit plugs disposed between the bit line and the cell pillars, but not between the bit line and the dummy pillars, wherein each of the cell pillars and the dummy pillars includes a core pattern, a channel pattern on a side surface of the core pattern, and an information storage pattern of an external side surface of the channel pattern.
10. The semiconductor device of claim 1, further comprising: a main pattern in the main trench; and a dummy pattern in the dummy trench, wherein each of the main pattern and the dummy pattern includes a conductive material line and an insulating spacer on a sidewall of the conductive material line.
11. A semiconductor device comprising: a substrate having a first surface and a second surface facing each other; a plurality of blocks on the first surface of the substrate and including a first block, a second block, and a third block; and a plurality of trenches including a first trench adjacent to the first block, a second trench adjacent to the second block, and a third trench adjacent to the third block, wherein the plurality of blocks are sequentially arranged in a first direction parallel to the first surface of the substrate, wherein the first block includes: first insulating layers and first conductive material layers, which are alternately and repeatedly stacked, and first pillars penetrating through the first insulating layers and the first conductive material layers in a vertical direction perpendicular to the first surface of the substrate, wherein the second block includes: second insulating layers and second conductive material layers, which are alternately and repeatedly stacked, and second pillars penetrating through the second insulating layers and the second conductive material layers in the vertical direction, wherein an uppermost first conductive material layer of the first conductive material layers and an uppermost second conductive material layer of the second conductive material layers are substantially at the same level, wherein an upper end of the first trench and an upper end of the second trench are at the higher level than the uppermost first conductive material layer, wherein a lowermost end of the first trench and a lowermost end of the second trench are at different levels from each other, wherein each of plurality of trenches extends in a second direction parallel to the first surface of the substrate, and wherein the second direction is perpendicular to the first direction.
12. The semiconductor device of claim 11, wherein the first trench and the second trench are sequentially arranged in the first direction, and wherein the lower end of the first trench is at a lower level than the lower end of the second trench.

13. The semiconductor device of claim 11, wherein the lower end of the first trench is at a lower level than a lowermost first conductive material layer of the first conductive material layers, and wherein the lower end of the second trench is at a lower level than a lowermost second conductive material layer of the second conductive material layers.
14. The semiconductor device of claim 11, wherein a width of the first trench in the first direction is greater than a width of the second trench in the first direction.
15. The semiconductor device of claim 11, further comprising: a first pattern in the first trench; and a second pattern in the second trench, wherein each of the first and second patterns includes a conductive material line and an insulating spacer on a sidewall of the conductive material line.
16. The semiconductor device of claim 11, wherein the third block includes: third insulating layers and third conductive material layers, which are alternately and repeatedly stacked, and third pillars penetrating through the third insulating layers and the third conductive material layers in the vertical direction, wherein an uppermost third conductive material layer of the third conductive material layers is substantially at the same level as the uppermost first conductive material layer of the first conductive material layers, wherein an upper end of the third trench is at the higher level than the uppermost first conductive material layer, wherein the first trench, the second trench and the third trench are sequentially arranged in the first direction, wherein the lower end of the first trench is at a lower level than a lowermost first conductive material layer of the first conductive material layers, wherein the lower end of the second trench is at a lower level than a lowermost second conductive material layer of the second conductive material layers, and wherein the lower end of the third trench is at a lower level than a lowermost third conductive material layer of the third conductive material layers.
17. A semiconductor device comprising: a substrate having a first surface and a second surface facing each other; a plurality of blocks on the first surface of the substrate and including a main block and a dummy block; a plurality of trenches including a main trench adjacent to the main block and a dummy trench adjacent to the dummy block, wherein the plurality of blocks are sequentially arranged in a first direction parallel to the first surface of the substrate, wherein the main block includes: main insulating layers and main conductive material layers, which are alternately and repeatedly stacked, and cell pillars penetrating through the main insulating layers and the main conductive material layers in a vertical direction perpendicular to the first surface of the substrate, wherein the dummy block includes: dummy insulating layers and dummy conductive material layers, which are alternately and repeatedly stacked; a lower mold layer between the substrate and a lowermost dummy conductive material layer of the dummy conductive material layers; and dummy pillars penetrating through the dummy insulating layers, the dummy conductive material layers, and the lower mold layer in the vertical direction wherein an uppermost main conductive material layer of the main conductive material layers is substantially at the same level as an uppermost dummy conductive material layer of the dummy conductive material layers, wherein an upper end of the main trench and an upper end of the dummy trench are at the higher level than the uppermost main conductive material layer, wherein a lowermost main conductive material layer of the main conductive material layers is substantially at the same level as the lower mold layer of the dummy block, and wherein a lowermost end of the main trench is at a lower level than a lowermost end of the dummy trench.
18. The semiconductor device of claim 17, wherein the lower mold layer extends between the first surface of the substrate and a lower surface of the dummy trench, and wherein the lower end of the main trench is at a lower level than the lower end of the dummy trench.
19. The semiconductor device of claim 17, wherein each of the main trench and the dummy trench extends in a second direction parallel to the first surface of the substrate, wherein the second direction is perpendicular to the first direction, wherein a width of the main trench in the first direction is greater than a width of the dummy trench in the first direction.
20. The semiconductor device of claim 17, further comprising: a main pattern in the main trench;

and a dummy pattern in the dummy trench, wherein each of the main pattern and the dummy pattern includes a conductive material line and an insulating spacer on a sidewall of the conductive material line.